

Brian Jia

(971) 901-8867

brian.jia04@gmail.com

ENGINEERING EXPERIENCE

International Rocket Engineering Competition Team, OSU

Avionics Lead

May 2025 – Apr 2026

- Designed guidance computer electrical schematics and PCBs; frequent use of electronic test equipment including oscilloscope, power supply, multimeter, VNA, etc. for validating embedded systems.
- Designed and validated low-noise DC-DC power converter circuits for powering IMUs, A/D & D/A converters, control surface motors, and low-noise transmission lines for GNSS.
- Wrote embedded C & C++ for ARM microcontrollers (RP2350, Cortex-M33) running FreeRTOS, using I2C, SPI, UART for sensor communications and flash memory interfaces.
- Conducted hardware-in-the-loop testing of student-made flight control surfaces with custom simulation plugin that generates synthetic sensor data and sends it to the flight computer to make it believe it is flying.
- Simulated and validated custom antennas and stock antenna placements for 433 MHz, 900 MHz bands.
- Leveraged Arduino and Raspberry Pi's education-focused ecosystems for rapid prototyping.
- Negotiated system-level trade-off analyses with other technical leads and executive management.

Co-Manager

May 2026 – Present

- Overseeing all aspects of engineering, logistics, and marketing for a student rocketry team building a two-stage experimental rocket targeting 45,000 ft.
- Reduced duplication of effort and increased team velocity by negotiating interfaces between sub-teams, e.g. by consolidating metal machining into a central manufacturing team.

EDUCATION

Ohio State University (OSU)

Expected Graduation: May 2027

- B.S. Electrical and Computer Engineering, *in progress*
- Honors Program, Maximus & Duhamel ECE Scholarship Recipient
- Overall GPA (4.0 Scale): 3.7

WORK EXPERIENCE

Vertiv

Computer Engineering Intern

May 2025 – Aug 2025, May 2026 – Aug 2026

- Fixed critical bug that prevented multiple Fortune 500 customers from factory resetting their devices.
- Automated build process for critical C/C++ tool with CMake, allowing 30+ variants to be built with a single configure/build step. Used LLVM/MinGW to enable cross-compilation of Windows binaries on Linux.

Center for Design & Manufacturing Excellence, OSU

Additive Manufacturing Research Assistant

Oct 2025 – Present

- Transforming a commercial metal 3D printer into a bespoke platform for additive manufacturing research.
- Leveraging Claude Code and OpenAI Codex to increase developer velocity while maintaining a high level of safety and robustness. Writing C# custom control plugins for Bosch Rexroth motor controllers,

Department of Engineering Education, OSU

Embedded Engineer / PCB Designer / Undergrad TA

Aug 2024 – Oct 2025

- Reduced costs by 32% (\$7000), enhanced manufacturability for large-scale (125 units) production for redesign of robot controller central to Honors Engineering curriculum,
- Developed electronics, PCBs, and student-friendly C/C++ robotics libraries, retaining API compatibility with last-generation robot controller.
- Validated design for resilience under misuse scenarios: GPIO & motor short circuit/overcurrent, overheating.